Part 12: Shifts

1. **What is the difference between logical and arithmetic shift?** Logical shift fills empty bits with 0. Arithmetic shift preserves the sign bit (MSB) to handle signed numbers correctly.
2. **How does SHL work? Shift Left**: It moves bits to the left. The LSB is filled with 0, and the MSB moves into the Carry Flag.
3. **How does SHR work? Shift Right**: It moves bits to the right. The MSB is filled with 0, and the LSB moves into the Carry Flag.
4. **How does SAL differ from SHL?** In 8086, `SAL` (Shift Arithmetic Left) and `SHL` are the **exact same instruction** and generate the same machine code.
5. **How does SAR work for signed numbers? Shift Arithmetic Right**: It shifts right but **copies the original Sign Bit** into the new vacant MSB position to keep the number positive/negative.
6. **How many bits can you shift in one instruction?** You can shift by **1** directly (e.g., `SHL AX, 1`) or by **CL** for multiple bits (e.g., `MOV CL, 3; SHL AX, CL`).
7. **How do shifts affect carry flag?** The last bit that is "shifted out" of the register falls into the **Carry Flag (CF)**.
8. **How do you multiply a number by 2 using shift?** Use `SHL` (Shift Left) by 1 position.
9. **How do you divide a number by 2 using shift?** Use `SHR` (Shift Right) for unsigned numbers, or `SAR` (Shift Arithmetic Right) for signed numbers.
10. **How do you shift multi-bit numbers in registers?** Shift the lower word first, then use `RCL` (Rotate through Carry) or `RCR` on the upper word to pull in the bit shifted out of the lower word.
11. **How do you implement bit masking using shift?** Shift the number to align the bits you want, then use `AND` to isolate them (masking out the others).
12. **How is ROL different from SHL? Rotate Left (ROL)** takes the bit falling off the left side and puts it back into the right side (circular). SHL just throws it away (into Carry).
13. **How is ROR different from SHR? Rotate Right (ROR)** takes the bit falling off the right side and puts it back into the left side. SHR fills with 0.
14. **Can you shift memory contents directly?** Yes. Example: `SHL BYTE PTR [BX], 1` shifts the byte at the memory address in BX.
15. **How do shifts help in bit manipulation?** They allow you to access specific bits, pack/unpack data, and perform very fast multiplication and division by powers of 2.